Set A1 5 December, 2023

July 2023 Term
CSE 471 – Machine Learning Class Test# 1
Full Marks: 20 Time: 20 minutes

Name: ____________________ Student No. ____________

No extra sheet shall be provided.

1. For the training set of the "WillWait" problem, calculate the information gain of the Patrons ("Pat") attribute **(10 marks)**

Example	Alt	Bar	Fri	Hun	Pat	Price	Rain	Res	Type	Est	WillWait
x_1	Yes	No	No	Yes	Some	$$$	No	Yes	French	0–10	$y_1 = Yes$
x_2	Yes	No	No	Yes	Full ✓	$	No	No	Thai	30–60	$y_2 = No$
x_3	No	Yes	No	No	Some	$	No	No	Burger	0–10	$y_3 = Yes$
x_4	Yes	No	Yes	Yes	Full ✓	$	Yes	No	Thai	10–30	$y_4 = Yes$
x_5	Yes	No	Yes	No	Full ✓	$$$	No	Yes	French	>60	$y_5 = No$
x_6	No	Yes	No	Yes	Some	$$	Yes	Yes	Italian	0–10	$y_6 = Yes$
x_7	No	Yes	No	No	None *	$	Yes	No	Burger	0–10	$y_7 = No$
x_8	No	No	No	Yes	Some	$$	Yes	Yes	Thai	0–10	$y_8 = Yes$
x_9	No	Yes	Yes	No	Full ✓	$	Yes	No	Burger	>60	$y_9 = No$
x_{10}	Yes	Yes	Yes	Yes	Full ✓	$$$	No	Yes	Italian	10–30	$y_{10} = No$
x_{11}	No	No	No	No	None	$	No	No	Thai	0–10	$y_{11} = No$
x_{12}	Yes	Yes	Yes	Yes	Full ✓	$	No	No	Burger	30–60	$y_{12} = Yes$

2. Calculate precision and recall based on the below confusion matrix. **(10 marks)**

Actual Class	Predicted Class: Churn=True	Predicted Class: Churn=False
Churn = True	15 TP	5 FN
Churn = False	9 FP	22 TN

Ans: no. 1

$N(yes) = 6, \quad N(No) = 6$

Entropy of the entire dataset $= -\frac{6}{12}\log_2\frac{6}{12} - \frac{6}{12}\log_2\frac{6}{12}$

$= 1$

Entropy of Pat attribute:

Entropy of Some $= -\frac{4}{4}\log\frac{4}{4} - \frac{0}{4}\log\frac{0}{4} = 0$

Entropy of Full $\{yes = 2, No = 4\} = -\frac{2}{6}\log\frac{2}{6} - \frac{4}{6}\log\frac{4}{6} = 0.918$

Entropy of None $\{yes = 0, No = 2\} = -\frac{0}{2}\log\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2} = 0$

$\therefore$ Information Gain = Entropy (whole data) $- \frac{4}{12}$ Ent (Some) $-$

$\frac{6}{12}$ Ent (Full) $- \frac{2}{12}$ Ent (None)

$= 1 - 0 - 0.918 - 0 = \boxed{0.082}$ Ans

Ans: no. 2

TP = 15, FN = 5, FP = 9, TN = 22

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{15}{15+9} = 0.625$$

$$\text{Recall} = \frac{TP}{P} = \frac{15}{15+5} = 0.75$$

Ans.

Set B1 5 December, 2023

July 2023 Term
CSE 471 – Machine Learning Class Test# 1
Full Marks: 20 Time: 20 minutes

Name: ______________________ Student No. ______________________

No extra sheet shall be provided.

1. Consider the following data frame where you are applying logistic regression. **(10 marks)**
    a. What is the current amount of loss?
    b. What is the gradient of loss w.r.t the feature f3 for the data point x5?

	f1	f2	f3	f4	f5	y	H	log(h)	log(1-h)
x1	0.85	0.41	0.28	0.35	0.92	1	0.25	-0.6	-0.12
x2	0	0.04	0.54	0.61	0.05	0	0.01	-2	0
x3	0.86	0.64	0.15	0.24	0.78	0	0.89	-0.05	-0.96
x4	0.75	0.18	0.71	0.56	0.35	1	0.91	-0.04	-1.05
x5	0.09	0.23	0.85	0.78	0.62	1	0.62	-0.21	-0.42
x6	0.11	0.66	0.3	0.73	0.04	1	0.78	-0.11	-0.66
x7	0.76	0.36	0.06	0.81	0.78	0	0.02	-1.7	-0.01
x8	0.33	1	0.17	0.68	0.99	1	0.75	-0.12	-0.6
x9	0.27	0.46	0.33	0.77	0.5	0	0.75	-0.12	-0.6
x10	0.82	0.1	0.88	0.67	0.85	0	0.2	-0.7	-0.1

2. Calculate precision and recall based on the below confusion matrix. **(10 marks)**

		Predicted Class	
		Churn=True	Churn=False
Actual Class	Churn = True	25	8
	Churn = False	6	12

Ans: no. 1 (a)

$$Loss = -\sum_{i=1}^{10}\left(y_i \log(h_i) + (1-y_i)\log(1-h_i)\right)$$

$$= -[1\times(-0.6) + 0\times(-.12) + 0\times(-2) + 1\times 0 + 0\times(-0.05) + 1\times(-0.96)$$

$$+ 1\times(-0.04) + 0\times(-1.05) + 1\times(-0.21) + 0\times(-0.42) +$$

$$1\times(-0.11) + 0\times(-0.66) + 0\times(-1.7) + 1\times(-0.01) +$$

$$1\times(-0.12) + 0\times(-0.6) + 0\times(-0.12) + 1\times(-0.6) +$$

$$0\times(-0.7) + 1\times(-0.1)]$$

$$= 2.75$$ Ans

Ans: no. 1(b)

$$L(y,h) = -\left(y\log h + (1-y)\log(1-h)\right)$$

$$\frac{\delta L}{\delta h} = -\frac{y}{h} + \frac{1-y}{1-h}$$

$$\frac{\delta L}{\delta f_3} = \frac{\delta L}{\delta h} \times \frac{\delta h}{\delta f_3} = \left(\frac{1-y}{1-h} - \frac{y}{h}\right) \times \omega_3$$

$$= \left(\frac{0}{1-0.62} - \frac{1}{0.62}\right) \times \omega_3$$

$$= -1.613\,\omega_3$$

Ans.

Corrected

$$h = \text{sigmoid}(z) = \frac{1}{1+e^{-z}}$$

$$\frac{\delta L}{\delta h} = \frac{\delta}{\delta h}\left(-y\log h - (1-y)\log(1-h)\right) = -\frac{y}{h} + \frac{1-y}{1-h}$$

$$\frac{\delta h}{\delta z} = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right) \Rightarrow h(1-h)$$

$$\frac{\delta z}{\delta f_3} = \omega_3$$ (চেইনরুল নাই, এখানেই লাগবে)

$$\therefore \frac{\delta L}{\delta f_3} = \frac{\delta L}{\delta h} * \frac{\delta h}{\delta z} * \frac{\delta z}{\delta f_3}$$

$$= \left(-\frac{y}{h} + \frac{1-y}{1-h}\right) \cdot h(1-h) \cdot \omega_3$$

$$= (h-y) \cdot \omega_3$$

$$= -0.38\,\omega_3$$

Set A2 **16 January, 2024**

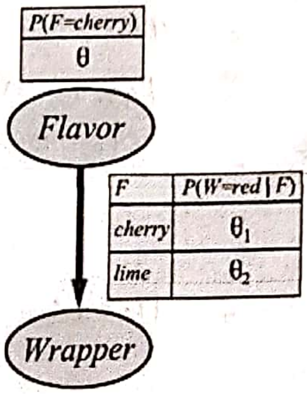
July 2023 Term
CSE 471 – Machine Learning Class Test# 2
Full Marks: 20 Time: 20 minutes

Name: ____________________ Student No. ____________________

No extra sheet shall be provided.

1. Consider the Bayesian Network for candies that may have cherry or lime flavor, and red or green wrappers. In a bag of candies, the count of various types of candies is as follows.

Flavor	Wrapper	
	Red	***Green***
Cherry	**30**	**20**
Lime	**25**	**15**



Write down the expression for the log likelihood of this bag. From it, deduce the values of θ, θ_1 and θ_2 that will maximize the likelihood of this bag. Use base 10 log in your calculations. **(10 marks)**

2. For the univariate data below, based on current parameter settings, the $p_{ij} = P(C = i|x_j)$ values have already been calculated in the **E-step** for **mixture of 2 Gaussians.** Calculate the model parameter values in the **M-step. (10 marks)**

Data Id	Feature	$P(C = 1\|x_j)$	$P(C=2\|x_j)$
x1	49	0.75	0.25
x2	53	0.5	0.5
x3	45	0.75	0.25
x4	45	0.8	0.2
x5	51	0.85	0.15

Ans: no. 2

$n_1 = 0.75 + 0.5 + 0.75 + 0.8 + 0.85 = 3.65$

$n_2 = 5 - 3.65 = 1.35$

$$\mu_1 = \frac{49 \times 0.75 + 53 \times 0.5 + 45 \times 0.75 + 45 \times 0.8 + 51 \times 0.85}{3.65}$$

$= 48.315$

$$\Sigma_1 = \frac{(49-48.315)^2 \times 0.75 + (53-48.315)^2 \times 0.5 + (45-48.315)^2 \times 0.75 + (45-48.315)^2 \times 0.8 + (51-48.315)^2 \times 0.85}{3.65}$$

$= 9.449$

$$w_1 = \frac{n_1}{N} = \frac{3.65}{5} = 0.73$$

$$w_2 = \frac{n_2}{N} = \frac{1.35}{5} = 0.27$$

$$\mu_2 = \frac{49\times0.25 + 53\times0.5 + 45\times0.25 + 45\times0.2 + 51\times0.15}{1.35}$$

$$= 49.37$$

$$\Sigma_2 = \frac{(49-49.37)^2\times0.25 + (53-49.37)^2\times0.5 + (45-49.37)^2\times0.25 + (45-49.37)^2\times0.2 + (51-49.37)^2\times0.15}{1.35}$$

$$= 11.566$$

Ans: no. 1

let, c – cherry, r_c – red wrapper, g_c – green wrapper
l – lime, r_l – red wrapper, g_l – green wrapper

$$P(d \mid h_{\theta,\theta_1,\theta_2}) = \theta^c (1-\theta)^l \cdot \theta_1^{r_c} (1-\theta_1)^{g_c} \cdot \theta_2^{r_l} (1-\theta_2)^{g_l}$$

$$L = [c \log\theta + l \log(1-\theta)] + [r_c \log\theta_1 + g_c \log(1-\theta_1)] + [r_l \log\theta_2 + g_l \log(1-\theta_2)]$$

$$\frac{\delta L}{\delta\theta} = \frac{c}{\theta} - \frac{l}{1-\theta} = 0 \Rightarrow \theta = \frac{c}{c+l} = \frac{50}{50+40} = 0.55$$

$$\frac{\delta L}{\delta\theta_1} = \frac{r_c}{\theta_1} - \frac{g_c}{1-\theta_1} = 0 \Rightarrow \theta_1 = \frac{r_c}{r_c+g_c} = \frac{30}{30+20} = 0.6$$

$$\frac{\delta L}{\delta\theta_2} = \frac{r_l}{\theta_2} - \frac{g_l}{1-\theta_2} = 0 \Rightarrow \theta_2 = \frac{r_l}{r_l+g_l} = \frac{25}{25+15} = 0.625$$

Ans.